

Jon Kirkpatrick

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Recent graduate of a joint pure mathematics and computer science program with strong academics and hands-on FPGA-based ML design experience. Leverages 15 years of technical, managerial, and precision-oriented work, now focused on computer science and applied research.

EDUCATION

Memorial University St John's, NL
B.Sc. *2023 – 2025*

- Pure Mathematics & Computer Science
- Dean's List (2024–2025)
- Cumulative GPA: 3.8/4.0
- Research in inference engine design (see Projects)

TECHNICAL EXPERIENCE

Quester Tangent Saanichton, BC
Production Assembler *2020 – 2022*

- Assembled and repaired high-density electronic boards, performing SMD soldering and diagnostic troubleshooting.
- Gained experience interpreting circuit schematics and applying precision repair techniques.

OTHER PROFESSIONAL EXPERIENCE

Autobody Industry Calgary & Victoria
Various Positions *2005–2020*

Over the course of fifteen years, I advanced from hands-on technical roles to senior management, developing strong problem-solving skills, attention to detail, and the ability to lead teams through complex projects. This experience required balancing precision work with customer-facing responsibilities, fostering adaptability and clear communication. While in a different industry, these capabilities continue to inform my approach to research, software development, and collaborative technical projects.

ACADEMIC PROJECTS

FPGA Inference Engine Research
Lead Developer *2025*

- Developed integer-only CNN engine for deployment on FPGA hardware, achieving near real-time inference.
- Designed unified convolution module capable of depthwise, pointwise, and fully connected layers.
- Developed vector dot product module capable of computing 8 bit by 256 element dot products every second clock cycle on constrained hardware.

Search Algorithms Coursework
Enthusiastic Student *2025*

- 1st place out of 80+ participants for both fastest BFS and A* implementations.
- Optimized runtime by focusing on constant time implementation and efficient queue management.

SKILLS

- **Programming Languages:**
Java, JavaScript, Python, C/C++, SQL
- **Frameworks/Libraries:**
PyTorch, Numpy, Pandas, scikit-learn
- **Tools & Environments:**
Git, Linux, Quartus, VS Code, IntelliJ, Eclipse
- **Areas of Interest:**
Algorithms, Neural Networks, Graph Theory